

Coding for AI

Multi-Agent Intelligence Hub

Name: NAVNEET DAS

Objective

This assignment focuses on managing structured data in Python through the use of sets, tuples, and nested dictionaries. To achieve this, a Multi-Agent Intelligence Hub (MAIH) is built to track various agents' locations and data. The system is designed to identify shared data across all agents, process agent movements, illustrate the immutable nature of tuples, and produce a summarized report of the data each agent holds.

Implementation

A nested dictionary named `agent_registry` is used to store the state of each agent. Every agent contains two attributes: `Location` and `Knowledge`. `Location` is represented using a tuple containing fixed integer coordinates, while `Knowledge` is represented using a set of information bits.

The common knowledge of all agents is found using set intersection. The function `find_common_intelligence()` collects the knowledge sets and calculates their intersection.

Summary Report

A dictionary comprehension is used to calculate the number of knowledge bits currently held by each agent.

```
summary_report = {
    agent_id: len(data["Knowledge"])
    for agent_id, data in agent_registry.items()
}
```

The resulting summary is:

```
{'Agent_Alpha': 3,
 'Agent_Beta': 3,
 'Agent_Gamma': 4}
```

TypeError and Tuple Immutability

When the program tries to change an element of the `Location` tuple directly, Python raises a `TypeError`. This happens because tuples are immutable, which means that their elements cannot be changed after the tuple has been created. In contrast, lists are mutable and their elements can be modified directly. For example, an individual element of a list can be replaced, whereas an individual element of a tuple cannot. Therefore, the program does not modify the existing location tuple. Instead, it creates a new tuple containing

the new coordinates and assigns it to the `Location` key. This demonstrates why tuples are suitable for representing fixed coordinate pairs. The error confirms that Python protects the contents of a tuple from direct item assignment.

Conclusion

Overall, the Multi-Agent Intelligence Hub serves as a practical application of core Python features, including tuples, sets, nested dictionaries, the `setdefault()` method, and dictionary comprehensions. The application effectively tracks agent movements, manages immutable data, pinpoints shared intelligence, and summarizes data distributions. Ultimately, these chosen data structures offer a highly efficient approach to organizing and manipulating structured data in Python.